DEEPFAKE X — FORENSIC MEDIA INTELLIGENCE REPORT

STRUCTURED FORENSIC EVIDENCE EXAMINATION & METADATA AUDIT

Investigation ID: DFX-STAGE2-FULL	**Date/Time:** 2026-11-12 08:00:00 UTC
Case Title: Comprehensive Stage 2 Forensic Investigation	**Investigator:** Lead Investigator
Overall Assessment: Conflicting forensic evidence	**Custody Standard:** Structured Forensic Evidence Handling

1. EXECUTIVE EVIDENTIARY ASSESSMENT

Forensic Category: Conflicting forensic evidence
Evidentiary Strength: Mixed / Divergent Cross-Modal Signals

Technical Rationale: Image flagged synthetic, while video and audio predict authentic.

2. EVIDENCE INVENTORY & CHAIN OF CUSTODY (SHA-256)

Filename	Modality	Size	SHA-256 Digest
surveillance.mp4	Video	2.4 MB	VID_SHA_ROOT
audio_track.wav	Audio	0.8 MB	AUD_SHA_CHILD

3. DETECTOR INFERENCE FINDINGS (UNCALIBRATED SOFTMAX SCORES)

Image Analysis (Swin Transformer Vision Backbone):
• **snapshot.png**: Model predicts AI-Generated Image | Synthetic: 91.5% | Authentic: 8.5% | Device: NVIDIA GeForce RTX 2050

Video Analysis (Uniform Frame Screening):
• **surveillance.mp4**: Flagged 1 / 6 frames synthetic | Mean Synthetic: 38.0% | Peak: 52.0%

Audio Analysis (Wav2Vec 2.0 Sequence Classifier):
• **audio_track.wav**: Model predicts Authentic Human Speech | Synthetic: 1.0% | Authentic: 99.0%

4. METADATA & CONTAINER COMPARISON MATRIX

Media Pair	Comparison Type	Agreement	Key Observation
surveillance.mp4 ↔ audio_track.wav	LINKED_AUDIO_VIDEO_STREAM_ANALYSIS	DISAGREEING_FINDINGS	Linked stream duration matches within tolerance.

5. GROUNDED FORENSIC EXPLAINABILITY & OPERATIONAL BOUNDS

• **Image 'snapshot.png':** Architecture: SwinForImageClassification | Score: Synthetic 91.5% / Authentic 8.5% | Decision: Model predicts AI-Generated Image
  Preprocessing: AutoImageProcessor ImageNet standard (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
  Raw Logits: Unavailable (raw neural logits are internal to the model execution context and not exported in the detector return dictionary)
  Key Limitation: Uncalibrated Softmax Activations: Probabilities reflect normalized exponential outputs of a discriminative linear head, not true frequentist probabilities of generation.
• **Video 'surveillance.mp4':** Sampled 6 frames (Uniform temporal spacing (step_msec = duration / requested_frames)) | Flagged Frames: 1 | Mean Synthetic: 38.0% | Peak: 52.0% at 00:02.00
  Raw Logits: Unavailable (raw neural logits are internal to the model execution context and not exported in the detector return dictionary)
  Key Limitation: Codec Compression Attenuation: H.264/H.265 temporal inter-frame prediction (P-frames, B-frames) and macroblock quantization attenuate high-frequency generative artifacts, suppressing synthetic scores.
• **Audio 'audio_track.wav':** Architecture: Wav2Vec 2.0 Sequence Classification (Wav2Vec2ForSequenceClassification) | Classification: Model predicts Authentic Human Speech (Synthetic: 1.0%, Authentic: 99.0%)
  Preprocessing: Polyphase resampled to 16000 Hz mono float32

Raw Logits: Unavailable (raw neural logits are internal to the model execution context and not exported in the detector return dictionary)
Key Limitation: Vocoder Domain Shift (Empirical False Negatives): The Wav2Vec2 backbone was trained on legacy TTS and vocoders (HiFi-GAN, WaveNet); empirical benchmarks demonstrate 0.0% synthetic scores (100% false negative) on modern flow-matching neural voice clones (ElevenLabs).

6. ADVANCED EVIDENCE COMPARISON & PROVENANCE LINEAGE

A. Cryptographic Evidence Lineage (Strict Provenance):
• **Child:** audio_track.wav (SHA: AUD_SHA_CHIL...) ➔ **Parent:** surveillance.mp4 | **Basis:** Verified SHA-256 Digest Match (VID_SHA_ROOT...)

B. Video Temporal Detector-Score Analysis:
• **surveillance.mp4:** Evaluated 6 frames | Mean: 38.0% | Std Dev: 6.76 pts | Range: [30.0% - 52.0%] | Threshold Crossings (>50%): 1
Notice: Temporal score variance does NOT prove localized splicing or video manipulation.

7. INVESTIGATION AUDIT TIMELINE ANALYTICS

Ledger Sequence Integrity: VALID (Cryptographically Intact) | **Total Events Logged:** 2 | **Investigation Span:** 60.0s

Seq	Action	Timestamp (UTC)	Details	Event Hash (Truncate
0	CASE_INITIALIZED	2026-11-12 08:00:00 UTC	Case initialized	5daa28f3960fc70b...
1	EVIDENCE_INGESTED	2026-11-12 08:01:00 UTC	Ingested media	06a9a3283b8307e2...

Cryptographic Integrity Notice: Hash-chain validation confirms that no audit entry was inserted, removed, or altered out of sequence. It does not verify host system clock accuracy.

8. FORENSIC METHODOLOGY & EMPIRICAL DETECTOR LIMITATIONS

Forensic Evidentiary Disclaimer: This report is generated strictly for investigative screening. Model scores reflect raw uncalibrated softmax activations of deep neural feature extractors and do not represent absolute mathematical truth.
1. **Image Detection (Swin Transformer):** Susceptible to false negatives on novel generative diffusion and flow architectures.
2. **Video Screening:** Performs uniform frame-level screening; does not constitute face-swap or lip-sync biometric proof.
3. **Audio Detection (Wav2Vec2):** Known empirical vulnerability to modern neural flow-matching speech synthesis (e.g., ElevenLabs). Evaluated accuracy is vocoder-dependent; false negatives occur on state-of-the-art voice clones.
4. **Cross-Modal Comparison:** Pairwise container comparison does not prove common origin or intentional manipulation without independent contextual corroborate.